Short remainder-only certificates for the greatest common divisor

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Abstract

We give a remainder-only straight-line derivation of the greatest common divisor of two positive integers using $O(\log a/\log\log a)$ remainder calls, uniformly for $a > b \geq 1$. Previously generated values may be retained and reused, but numerical constants and other arithmetic operations are not available as instructions. The construction combines a stored Euclidean prefix with three-call contractions and two smaller recursive derivations. It yields the same upper bound for the intrinsic remainder-call complexity of the gcd and of coprimeness in the Euclidean structure. This refutes the uniform-process form of the conjecture of van den Dries and Moschovakis that the Euclidean algorithm is optimal among its peers: Conjecture 3.4 of Moschovakis's 2012 notes, and the precise version (4F-5) of the Main Conjecture in his book. It does not give a deterministic remainder-only algorithm with this cost, and the deterministic conjecture remains open. All results are formalized in Lean 4; the final section states exactly what the formalization covers.

Statement and relation to the book

Throughout, logarithms are to base 2. A derivation starts with its two inputs. An instruction adds $\operatorname{rem}(x,y)$, where x, y are already available and $y > 0$. Earlier values remain available. No constants are supplied. Let $G(a,b)$ be the minimum number of instructions needed to make $\gcd(a,b)$ available, and let $D(a,b) = G(a,b)$ for coprime inputs. Thus $G(a,0) = 0$ and $D(a,1) = 0$. These are costs of generating values.

Theorem 1. *Uniformly for integers $a > b \geq 1$,*

$$G(a,b) = O\left(\frac{\log a}{\log\log a}\right).$$

In particular this bound holds for $D(a,b)$ when $\gcd(a,b) = 1$.

As usual, the asymptotic assertion concerns sufficiently large a; a bound valid at every positive input is $O(1 + \log(2+a)/\log\log(4+a))$.

Background.

Van den Dries and Moschovakis asked whether the Euclidean algorithm is optimal among its peers, the algorithms that compute the gcd from the remainder function [1]. Moschovakis's Main Conjecture asserts that every algorithm from $\operatorname{rem}, \operatorname{eq}_0$ computing $\gcd(x,y)$ makes more than $r \log x$ remainder calls on infinitely many pairs [2, p. 2]. The known lower bound is a logarithm smaller. A recursive program deciding coprimeness, even from $=, <, +$, truncated subtraction, iq and rem, makes more than $\frac{1}{10}\log\log a$ calls on infinitely many coprime pairs [1]; see also [3] and [2, Theorem 6C.5]. This bound holds for all uniform processes, nondeterministic ones included. Pratt's nondeterministic algorithm attains it on successive Fibonacci numbers [2, Theorem 2E.2, Corollary 6C.7]. Whether a logarithmic bound holds for all uniform processes was left open [2, p. 188], [4, Conjecture 3.4].

In a different model, Bshouty showed that a random access machine with linear operations, shifts and indirect addressing computes the gcd in $O(\ \log N/\log\ \log N\ )$ operations, by reading precomputed blocks of the binary gcd algorithm from a table [5]. The construction here uses no operation other than remainder and no table. Its saving of a factor $\log\ \log a$ comes instead from reusing one stored Euclidean prefix for every modulus.

The conjectures refuted.

The definitions of generated certificates and intrinsic complexities are in [2, secs. 1D, 4C–4E] and are recalled in Section 5. Write $N_\varepsilon = \left(\ \mathbb{N},\ \mathrm{rem}\ ,\ \mathrm{eq}_0\ ,\ \mathrm{eq}_1\ \right)$ and $\mathrm{calls}_{\mathrm{rem}}\ \left(\ A, f, \vec{x}\ \right)$ for the least number of remainder facts in a certificate for f at $\vec{x}$ in A. Moschovakis's 2012 notes state:

> **Conjecture 3.4** [4]. There is a rational number $r > 0$ such that for infinitely many $a \geq b \geq 1$, $\mathrm{calls}_{\mathrm{rem}}\ (\ N_\varepsilon,\ \perp\ \perp\ , a, b\ ) \geq r \log a$.

The book formulates one precise version of the Main Conjecture, display (4F-5) [2, p. 151], as the existence of an infinite set D of pairs and a $K > 0$ such that

$$(\ x, y\ ) \in D\ \Rightarrow\ 2 \log\ \max\ (\ x, y\ ) \leq K\, \mathrm{calls}_{\mathrm{rem}}\ (\ A,\ \gcd\ , x, y\ ).$$

Corollary 2. *The intrinsic complexities* $\mathrm{calls}_{\mathrm{rem}}\ (\ N_\varepsilon,\ \perp\ \perp\ , a, b\ )$ *and* $\mathrm{calls}_{\mathrm{rem}}\ (\ \mathcal{N}_0,\ \gcd\ , a, b\ )$ *are* $O(\ \log a/\log\ \log a\ )$*, uniformly for* $a \geq b \geq 1$*. Consequently Conjecture 3.4 of [4] is false, and so is the version (4F-5) of the Main Conjecture: for every* $r > 0$*, respectively* $K > 0$*, only finitely many pairs satisfy the stated inequality.*

The corollary follows from Theorem 1 and Section 5. It is formalized as `ShortGCD.not_conjecture_3_4` and `ShortGCD.not_main_conjecture_4F5`. The structure A in (4F-5) may include eq_1 ; allowing more kinds of facts can only lower the complexity, so the refutation applies either way. Theorem 4F.1 [2, pp. 149–150] transfers these pointwise intrinsic bounds to a single uniform process.

The corollary settles neither the Main Conjecture for deterministic algorithms nor its form for deterministic uniform processes [4, Conjecture 3.9], [2, sec. 4H]; see Section 6. With the lower bound above, the intrinsic complexity of coprimeness in N_ε now lies between $\frac{1}{10}\log\ \log a$, on infinitely many pairs, and $O(\ \log a/\log\ \log a\ )$, on all pairs. Closing this gap is an open problem.

Normalization

Write $g = \gcd\ (\ a, b\ ) > 0$. For positive y,

$$\mathrm{rem}\ (\ gx, gy\ ) = g\, \mathrm{rem}\ (\ x, y\ ).$$

Every generated value is divisible by g. Consequently scaling a derivation, or dividing all its values by g, preserves its length and the validity of every instruction. Hence

$$G\ (\ a, b\ ) = D\ (\ a/g, b/g\ ).$$

The normalized pair $(\ 1, 0\ )$ is understood to have cost zero. Define

$$T(\ n\ ) = \max_{0 \leq b \leq a \leq 2^n,\ a > 0} G\ (\ a, b\ ).$$

Euclid's algorithm makes this a finite maximum. Normalization reduces an upper bound for T to coprime inputs, with equality, zero inputs, and inputs equal to one handled without calls.

Three-call contractions

Lemma 3. *If* $m > 0$ *is available and a finite collection* S *of available values occupies* $t \geq 2$ *residue classes modulo* m*, at most three remainder calls generate* w *with*

$$0 < w < \frac{m}{t-1}.$$

Proof. Sort the distinct residues. Some adjacent pair $u < v$ has gap $\delta = v - u < m/(t-1)$, because the sum of the $t-1$ adjacent gaps is less than m. Each selected residue can be generated with one call from a witness in S. If $u = 0$, use v. If $0 < u \le \delta$, use u. Otherwise $0 < \delta < u$ and $v = u + \delta$, so $\operatorname{rem}(v, u) = \delta$. This uses at most three calls. Subtraction and sorting describe the proof and the selection of instructions; they are not instructions of the derivation. □

Lemma 4 (Euclidean-prefix lemma). *Let $r_0 = a > r_1 = b > \cdots > r_h > 0$ be a Euclidean prefix of a coprime pair, with $r_{i-1} = q_i r_i + r_{i+1}$ and $q_i \ge 1$. For every integer $m > 2a/r_h$, the h pairs*

$$(r_i \bmod m, r_{i+1} \bmod m), \qquad 0 \le i < h,$$

are distinct. The prefix therefore occupies t residue classes with $t^2 \ge h$.

Proof. Suppose the pairs at $i < j$ coincide. Set

$$M = \prod_{\ell=i+1}^{j} \begin{pmatrix} q_\ell & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} A & B \\ C & E \end{pmatrix}, \qquad \begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = M \begin{pmatrix} r_j \\ r_{j+1} \end{pmatrix}.$$

Right multiplication by $\left(\begin{smallmatrix} q & 1 \\ 1 & 0 \end{smallmatrix}\right)$ sends (A, B, C, E) to $(qA + B,\ A,\ qC + E,\ C)$, starting from $(q, 1, 1, 0)$ for a single factor. By induction, all entries are nonnegative, $A \ge B \ge E$, $A \ge C \ge 1$, $B \ge 1$, and $\det M = (-1)^{j-i}$. Also $Ar_j \le r_i \le a$, whence $A \le a/r_h$. Write

$$\Delta = \det(M - I) = 1 + \det M - A - E.$$

For an odd number of factors, $\Delta = -A - E < 0$. By the same recursion, a product of at least two factors has $A \ge 2$ and $E \ge 1$; thus for an even positive number of factors, $\Delta = 2 - A - E < 0$. In either case,

$$0 < |\Delta| \le A + E \le 2A \le 2a/r_h < m.$$

The presumed congruence gives $(M - I)\left(r_j, r_{j+1}\right)^{\mathsf{T}} \equiv 0 \pmod m$. Multiplication by the adjugate shows that m divides both Δr_j and Δr_{j+1}. Consecutive remainders are coprime, so Bézout's identity implies $m \mid \Delta$, a contradiction. Finally, t individual residue classes give at most t^2 ordered pairs. □

Thus a stored prefix can be reused for every available modulus $m > 2a/r_h$. When $\sqrt{h} > 1$, three calls suffice to generate

$$0 < w < \frac{m}{\sqrt{h} - 1}.$$

The universal quantifier over m is essential: the modulus generated by a contraction is not predetermined.

Construction and recurrence

Fix a sufficiently large integer n and put $N = 2^n$ and $h = \lceil \sqrt{n} \rceil$. After normalization we may assume $a > b > 1$ and $a \le N$. Generate the Euclidean prefix until its last positive value is at most $N^{3/4}$ or until r_h is available. Check the input b as part of this rule. The preliminary work costs at most h calls. Euclid cannot encounter zero first without having encountered the positive remainder 1.

If a positive remainder $c \le N^{3/4}$ is encountered, generate the next remainder $s < c$ with one more call. The pair (c, s) is coprime. If $s = 0$, then $c = 1$ is already available; otherwise both inputs to the remaining derivation are at most $2^{\lceil 3n/4 \rceil}$. This gives the bound

$$h + 1 + T(\lceil 3n/4 \rceil).$$

Otherwise $r_h > N^{3/4}$, so $2a/r_h < 2N^{1/4}$. For $n > 12$, $N^{1/3} > 2N^{1/4}$, and Lemma 4 applies to every available $m > N^{1/3}$. Starting at $m = r_h$, repeatedly use the contraction inequality until $0 < m \le N^{1/3}$. Set

$$\kappa_n = \sqrt{h} - 1 > 1, \qquad J_n = \left\lceil \frac{2n/3}{\log \kappa_n} \right\rceil .$$

There are at most J_n contractions: if J_n have occurred then $m < N/\kappa_n^{J_n} \le N^{1/3}$. Since $\log \kappa_n = \frac{1}{4} \log n + O(1)$, their cost is $3J_n = O(n/\log n)$.

The small m need not be coprime to either input. Generate $u = \operatorname{rem}(a, m)$ and $v = \operatorname{rem}(b, m)$, then concatenate derivations of $d = \gcd(m, u)$ and $e = \gcd(d, v)$. These derivations exist by the definition of T and use only inputs bounded by m. They are not gcd-oracle instructions. The gcd identities give

$$e = \gcd(\gcd(m, a \bmod m), b \bmod m) = \gcd(m, \gcd(a, b)) = 1.$$

The cost in this case is at most

$$h + 3J_n + 2 + 2T(\lceil n/3 \rceil).$$

Retaining values makes concatenation legitimate. In particular, the second recursive derivation has both d and v available even if the first one uses neither original input.

The two case estimates and normalization imply

$$T(n) \le \max\{T(\lceil 3n/4 \rceil) + O\left(\sqrt{n}\right),\ 2T(\lceil n/3 \rceil) + O(n/\log n)\}.$$

All implied constants are independent of the inputs.

Proof of Theorem 1. For $n \ge 2$ let $f(n) = n/\log n$. The ratios

$$\frac{f(\lceil 3n/4 \rceil)}{f(n)} \to \frac{3}{4}, \qquad \frac{2f(\lceil n/3 \rceil)}{f(n)} \to \frac{2}{3}$$

show that, beyond a fixed threshold, the two recursive terms have potentials at most $\frac{7}{8} f(n)$ and $\frac{5}{6} f(n)$, respectively. Both overheads are at most $Af(n)$ for a fixed A. Choose $C \ge 8A$ and increase C to cover the finitely many initial values with indices at least 2. Choose the induction threshold large enough that both recursive indices are at least 2 and strictly less than n. Strong induction applied to this recurrence then gives $T(n) \le Cf(n)$: the two bounds are $(7C/8 + A) f(n)$ and $(5C/6 + A) f(n)$, both at most $Cf(n)$. Finally take $n = \lceil \log a \rceil$. This also gives $G(a, b) = o(\log a)$ uniformly as $a \to \infty$. □

Certificates and uniform processes

Use the book's partial remainder operation: its second argument must be nonzero. Write $\mathcal{N}_0 = (\mathbb{N}, \operatorname{rem}, \operatorname{eq}_0)$ and $\mathcal{N}_\varepsilon = (\mathbb{N}, \operatorname{rem}, \operatorname{eq}_0, \operatorname{eq}_1)$; compare [2, pp. (1C–10), pp. 24, 142–143]. A derivation determines a finite generated partial substructure by recording its remainder equations. Repeated equations can be discarded.

For the relation $\gcd(a, b) = 1$ on all natural inputs, a derivation of 1 together with $\operatorname{eq}_1(1) = \mathrm{tt}$ certifies the relation: every common divisor of the images of the inputs divides every transported generated value, hence divides 1. Conversely, without a positive eq_1 fact the map $x \mapsto 2x$ preserves every recorded remainder and zero-test fact, as well as every negative one-test fact. It cannot preserve coprimeness. A certificate therefore reaches 1. Enumerating a finite generated diagram in generation order gives a derivation using no more calls than the number of remainder facts. Consequently, on coprime inputs,

$$\operatorname{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd = 1, a, b) = D(a, b), \qquad \operatorname{calls}(\mathcal{N}_\varepsilon, \gcd = 1, a, b) = D(a, b) + 1.$$

If the relation is restricted to positive inputs, $a > 0 \wedge b > 0 \wedge \gcd\,(a, b) = 1$, the certificate also needs the facts $\mathrm{eq}_0\,(a) = \mathrm{ff}$ and $\mathrm{eq}_0\,(b) = \mathrm{ff}$. The remainder count is unchanged, and the total count is at most $D(a, b) + 3$.

For gcd as a function, the certificate must reach its output g, giving a lower bound of $G(a, b)$. For an upper bound, append

$$\mathrm{rem}\,(a, g) = z, \qquad \mathrm{rem}\,(b, g) = z, \qquad \mathrm{eq}_0\,(z) = \mathrm{tt},$$

where the generated z is zero. Every homomorphism π has $\pi(z) = 0$ and $\pi(g) > 0$ (by definedness of either appended remainder). Thus $\pi(g)$ divides both input images. Conversely every common divisor of those images divides $\pi(g)$ by induction over the derivation. Therefore

$$G(a, b) \le \mathrm{calls}_{\mathrm{rem}}\,(\mathcal{N}_0, \gcd, a, b) \le G(a, b) + 2.$$

Positivity facts may be appended if the function is restricted to positive inputs. For a noncoprime input, add $\mathrm{eq}_1\,(g) = \mathrm{ff}$ to this gcd certificate to certify noncoprimeness. Thus the sublogarithmic remainder-call bound holds for both truth values of coprimeness as well as for gcd. Theorem 4F.1 supplies a single uniform process realizing each intrinsic norm pointwise [2, Theorem 4F.1].

Scope and verification

Selecting a close pair of residues is uncharged existential choice in a straight-line derivation. A deterministic remainder-only algorithm cannot assume that choice is available at constant cost. Computing all prefix residues would take about h calls per contraction. The construction therefore does not settle the deterministic conjecture.

The supplied offline Python planner uses unrestricted arithmetic to select instructions; its independent replay checks operand availability, nonzero divisors, every remainder equation, and the final gcd. On one pseudorandom coprime pair of each size, generated from its default seed, it gives:

Input bits	Derivation calls	Euclid calls to reach 1, same pair
512	196	326
2048	669	1204
8192	2059	4754

Random pairs are not Euclid's worst case: on successive Fibonacci numbers Euclid needs about $1.44 \log a$ calls. These are checks of finite instances, not an asymptotic proof. A reproducible suite of 310 cases also passes, including 175 noncoprime pairs, with 3,543 contractions across 335 reservoir phases, and four deliberately invalid traces are rejected. The Euclidean-prefix lemma was also checked exhaustively for every coprime $a < 1500$, every prefix length, and 60 moduli above the threshold $2a/r_h$ (254,722,179 instances, no failures).

Machine-checked statements

The Lean 4.35.0 development, with mathlib pinned to version 4.35.0, proves the uniform real-logarithmic bound, its worst-case Big-O formulation, the Euclidean-prefix lemma, the complete three-call contraction lemma, normalization, and the certificate upper bounds. It contains no proof placeholders or added axioms. An axiom audit of the main results reports only the standard Lean foundations `propext`, `Classical.choice`, and `Quot.sound`. The theorem `ShortGCD.uniform_real_bound` states

$$\exists C > 0\ \forall a, b \in \mathbb{N}, \quad 4 \le a \wedge b \le a \Rightarrow G(a, b) \le C \frac{\log_2 a}{\log_2\left(\log_2 a\right)}.$$

The proof starts from a finite-store instruction semantics in which the only instruction is remainder with a positive, available divisor. Costs are indices of derivations, rather than an assumed recurrence. A formal conversion to finite generated diagrams proves the gcd certificate bound with two extra remainder facts and the bound for positive and negative coprimeness instances. The equality with the existing D definition is also checked.

The intrinsic complexities themselves are also defined in Lean, directly from certificates: `callsCoprimeness` for the coprimeness relation in N_ε and `callsGcd` for the gcd without eq_1 facts. On coprime pairs the former agrees with the existing reduction to D (`callsCoprimeness_eq_callsRem`). The theorems `ShortGCD.not_conjecture_3_4` and `ShortGCD.not_main_conjecture_4F5` state that no $r > 0$ satisfies $\mathrm{calls}\,(\,\mathrm{rem}\,)\,(\,N_\varepsilon,\ \bot\ \bot\,,a,b\,) \geq r\log_2 a$ for infinitely many $a \geq b \geq 1$, and that no $K > 0$ satisfies $2\log_2\ \max\ (\,x,y\,) \leq K\,\mathrm{calls}\,(\,\mathrm{rem}\,)\,(\ \gcd\,,x,y\,)$ for infinitely many pairs. The general theory of uniform processes and Theorem 4F.1 are cited from the book, not re-formalized here. They are needed only to interpret these intrinsic bounds as statements about a single best uniform process, not for the refutations themselves.

Integer parameters used in Lean

To avoid real rounding in the construction itself, the formal proof uses, for $j = \lfloor \log_2 n \rfloor \geq 32$,

$$p = \lfloor j/4 \rfloor, \quad k = 2^p, \quad h = 2^{2p} + 1, \quad J = \lfloor n/p \rfloor + 1, \quad \ell\ = \lceil 3n/4 \rceil, \quad s = \lceil n/3 \rceil.$$

The finite construction proves, for arbitrary natural parameters satisfying $k \geq 1$, $k^2 < h$, $L \geq 1$, $2A \leq ML$, and $A \leq Mk^J$,

$$S\,(\,A\,)\ \leq\ \max\,\{h+1+S\,(\,L\,)\,,\ h+3J+2+2S\,(\,M\,)\,\},\qquad S\,(\,A\,)\ =\ \max_{a,b\,\leq\,A}\,G\,(\,a,b\,)\,.$$

Taking $A = 2^n, L = 2^\ell, M = 2^s$ gives the required hypotheses. Lean checks

$$(\,h+1\,)\,j \leq 64n, \quad (\,h+3J+2\,)\,j \leq 64n, \quad j-2 \leq\ \log_2^{\mathrm{nat}}\ \ell\,,\ \log_2^{\mathrm{nat}}\ s, \quad 8\,\ell\ \leq 7n, \quad 16s \leq 7n.$$

The last estimates and $8j \leq 9\,(\,j-2\,)$ give a factor $63/64$ for either recursive contribution after clearing denominators. Taking $C \geq 4096$ and large enough for the finite initial segment proves $S\,(\,2^n\,)\ \lfloor \log_2 n \rfloor \leq Cn$ for all n. Comparing integer floors and ceilings with real base-two logarithms yields the stated real bound. These parameter choices change only absolute constants, not the proof mechanism or its asymptotic order.

References

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